
AN AGENT-SUPERVISED LEARNED RESIDUAL ON MODEL PREDICTIVE PITCH CONTROL OF A WIND TURBINE

THE RESIDUAL ADDS TO THE STRONGEST CONTROLLER AND REPAIRS ITS MODEL; A
LANGUAGE-MODEL AGENT WRITES ITS REWARD

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ABSTRACT

Model predictive control (MPC) of wind-turbine pitch is only as good as its model, and a learned controller is only as good as its reward. We combine the two on collective-pitch control of the NREL 5 MW turbine in OpenFAST: a linear-parameter-varying MPC, which with a correctly scaled cost is the strongest controller on the metric set of a published reinforcement-learning pitch-control study (power MSE, generator-speed MSE, tower-base and blade-root fatigue), carries a bounded residual policy trained by PPO, and a language-model agent supervises the residual’s reward during training—every 30 episodes it writes candidate reward expressions, each is verified by a 30-episode fork against the objective, the best continues, and a guardrail rolls back. On two disjoint held-out wind sets the supervised residual adds 1.0 and 1.6 points to the exact-model MPC with all four metrics improved in every seed, and when the MPC’s power and thrust coefficients are 5 % wrong in its model it restores the controller from 8.5 / 3.6 to 12.7 / 18.3, above the exact-model MPC on one set: the residual learns what the model gets wrong. Ablations delimit the agent’s part. A fixed, tuned reward reaches a level within seed noise of the agent’s on the MPC base; what the agent demonstrably changes is the reward’s structure, visible on the weaker gain-scheduled-PI base where the residual has more to do: its rewards reach a balanced solution with every metric improved, including tower fatigue, in 3 of 5 seeds where no fixed-reward or weight-searched run reaches it in any, the agent asked once without verification does not, and a random search over the agent’s own vocabulary of term shapes, verified by the same loop, does. Every MPC-alone number is the MPC run as the residual’s base controller, through the same code path as the residual runs.

1 Introduction

Above rated wind speed a turbine’s collective pitch regulates rotor speed and power while limiting fatigue of the tower and the blades. Model predictive control (MPC) handles the trade-off explicitly and is a standard reference for the problem [1–5]; reinforcement learning (RL) has been proposed as a model-free alternative [6–9]. Each has a dependency the other does not: the MPC depends on its model, the learned controller on its reward. Residual learning addresses the first—a bounded policy on top of the model-based controller can learn what the model gets wrong—and language-model (LLM) agents are the newest answer to the second: Eureka evolves reward code against a task score [10], Language-to-Rewards and Text2Reward map instructions to reward parameters and programs [11, 12], DrEureka adds domain-randomisation design [13], and agentic reward designers close the loop with semantic feedback [14].

This paper puts the two together on one plant and measures what each part is worth. The controller is an LPV-MPC of the NREL 5 MW turbine whose cost is scaled so that its tower term works, which makes it the strongest controller we know of on the objective; a bounded PPO residual sits on its target; a reasoning-model agent supervises the residual’s reward during training under fork verification against the objective. The objective is the metric set of a published RL pitch-control study [9]—power MSE, generator-speed MSE, tower-base and blade-root fatigue, each as a paired percentage reduction against the gain-scheduled PI (GSPI)—and every number is a held-out evaluation on two disjoint wind sets with seed pairing and exact permutation tests.

Results. The supervised residual adds +1.0/+1.6 points to the exact-model MPC on the two held-out sets, above the MPC in five of six seed-set pairs, with all four metrics improved in every seed. When the MPC’s power and thrust coefficients are 5 % low in its internal model—the controller alone falls from 16.3/17.2 to 8.5/3.6—the residual restores 12.7/18.3, above the exact-model MPC alone on one set. The agent’s rewards share a structure no weight of a conventional reward can express: saturated speed and power terms on the above-rated subset, and bounded fatigue terms.

Ablations. A fixed, tuned reward on the same MPC base reaches a level within seed noise of the agent’s (+0.3/+0.8 over the MPC; difference to the agent +0.6/+0.8, two of three seeds, $n = 3$), so the agent’s contribution is not a level effect on this base. It is a structural one, which shows where the residual has more to do: on the GSPI base the agent’s rewards reach a balanced solution—every metric improved, tower fatigue included—in 3 of 5 seeds where no fixed-reward or weight-searched run reaches it in any; the agent asked once and applied without verification is the fixed reward; and a random search over the agent’s own vocabulary of term shapes, verified by the same loop, reaches the same balanced solution. The agent supplies the vocabulary; verification against the objective makes it real.

Contributions. (a) An MPC with a supervised learned residual, evaluated on held-out winds: it adds to the strongest controller on the objective and repairs a model error. (b) A corrected model-based reference: the MPC must be evaluated as the residual’s base controller, not through the learner’s action channel, and with $O(1)$ cost terms it is positive on every metric where the published verdict has it losing tower fatigue. (c) Ablations with matched controls that delimit the agent’s role to the structure of the reward.

2 Setup

2.1 Plant, objective, statistics

NREL 5 MW onshore turbine [15] in OpenFAST 4.2.1 [16], controlled by ROSCO 2.10.5 [17] through a patched ZeroMQ interface at the 10 ms control step; the ROSCO configuration is the gain-scheduled PI (GSPI) with peak shaving on and every other load-mitigation feature off, constant torque above rated, and it is the pairing reference for every metric. Wind: TurbSim, 8, 12.5 and 15 m/s mean hub wind, turbulence intensity 8 %, 150 s episodes of which 130 s are scored. Realisations with TurbSim seeds 1–2 are the supervisor’s evaluation winds; seeds 3–6 and 7–10 are two disjoint held-out sets (“wind seeds 3–6” and “wind seeds 7–10”). Fatigue is the damage-equivalent load from rainflow counting [18, 19]. The objective is the metric set of [9] as one scalar,

$$J = \frac{1}{4} \sum_{m \in \{P, \omega, T, B\}} \text{clip}_{\pm 100}(\Delta m [\%]) - 20 \max(0, E_{\text{loss}}[\%] - 1), \quad (1)$$

the mean of the per-episode-clipped percentage reductions, relative to paired GSPI, of power MSE and generator-speed MSE on above-rated steps (labelled by hub wind, whatever the controller does) and of tower-base and blade-root fatigue, minus a penalty for energy loss beyond 1 %. Every selection during training (fork winner, best checkpoint, rollback) uses J on wind seeds 1–2; every reported number is a held-out evaluation of the best checkpoint. Seeds are paired across arms; differences are reported as mean, wins/ n and the exact two-sided sign-flip permutation p (floor $2/2^n$: 0.0625 at $n = 5$, 0.25 at $n = 3$). A seed is *balanced* when all four metrics improve on a held-out set.

2.2 The model-based controller and what it depends on

A three-state LPV-MPC (rotor, first tower fore-aft mode) linearised every 0.1 s from the C_P/C_T tables, horizon 2 s, OSQP [20] (3.6 ms per solve), pitch within the live peak-shaving floor and the rate limit, wind held at ROSCO’s low-passed estimate (no preview), torque native. Its cost has three terms—speed error, tower-top velocity, pitch increment—each scaled to $O(1)$ at its typical value, and its weights are selected on wind seeds 1–2 by J , the same rule as every learned arm. The scaling is what makes it strong: with the speed error normalised by rated speed instead, the speed term is $O(10^{-5})$ at a typical error of 0.5 % while the tower term is $O(1)$, any tower weight out-weighs regulation by four orders of magnitude, the optimum feathers the rotor, and the best such MPC has no tower term and loses tower fatigue (−22 %)—the published verdict. With $O(1)$ terms the same MPC is positive on every metric: 16.3/17.2 on the two held-out sets.

That lead is conditional on the model (Fig. 1): tower frequency ± 10 % and modal mass −20 % cost 3–4 points, but the power and thrust coefficients 5 % low or high take the controller to 8.5 and 5.7, and ± 15 % to −24, its regulation collapsing. An identified aerodynamic model will not be exact. This is what a learned residual has to repair, and the 5 % case is the one we train on.

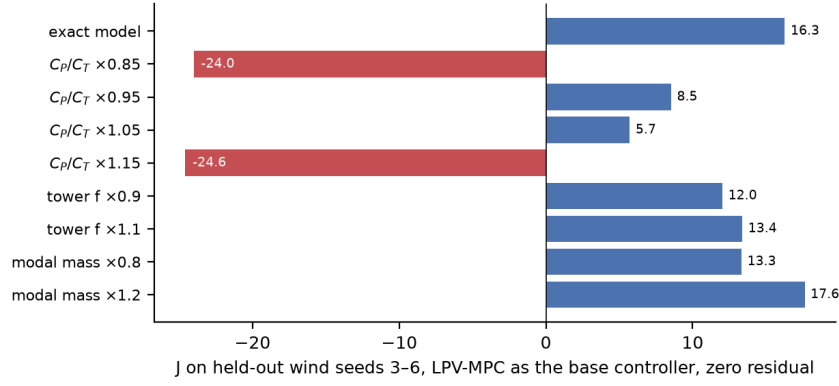


Figure 1: The MPC is conditional on its aerodynamic model. Held-out J (wind seeds 3–6) of the MPC when only its internal model is perturbed; the plant is unchanged. Errors on the tower model cost a few points; errors on the power and thrust coefficients cost the controller.

How the MPC is evaluated. Every MPC-alone number in this paper is the MPC run as the residual’s base controller with a zero residual, i.e. through exactly the code path of the residual runs. An earlier version sent its command through the residual channel of the RL agents instead; that channel carries a rule written for the learned residual—below rated, by the router’s region, the offset is non-negative—which zeroed the MPC’s negative offsets during region transitions at 12.5 m/s and produced pitch discontinuities: on one held-out episode the tower-base fatigue was 11.6 MN m through the channel and 7.3 MN m for the same MPC as the base controller, against 19.4 for GSPI; on the held-out sets the difference was four points of J . A reference controller has to be evaluated as itself.

2.3 The learned residual

Two PPO [21] agents—one per operating region, selected by an oracle rule on ROSCO’s native pitch command—add a bounded collective-pitch offset ($|\Delta\beta| \leq 0.05$ rad, second-order damped) to the MPC’s target; the pitch limits apply to the sum. Observations: generator-speed error and its rate, measured pitch, hub wind speed, blade-root out-of-plane moment, tower-top fore-aft acceleration, and the MPC’s target. $\gamma = 0.998$, GAE $\lambda = 0.98$ [22], 64×64 tanh networks, 300 training episodes per run; the critic’s regression target is standardised by a running estimate of the return scale [23]. With zero residual the controller is the MPC itself, so the episode-0 policy is the MPC and the best checkpoint is selected against it. The fixed reward used as a control is region-conditional,

$$r_t = \mathbf{1}[\text{Region 2}] w_P \left(\frac{P}{P_{\text{GSPI}}} - 1 \right) - \mathbf{1}[v_{\text{hub}} > v_{\text{rated}}] w_\omega \left(\frac{\delta\omega}{0.005} \right)^2 - \lambda_T \ell_T - \lambda_B \ell_B - \lambda_A (\Delta\beta/\kappa)^2, \quad (2)$$

with fatigue proxies ℓ_T, ℓ_B (growth of the 10 s peak-to-peak range of tower acceleration and of blade-root moment, ≈ 1 per step under GSPI) and the speed term gated by the wind label, the subset on which the objective’s regulation metrics are scored; its six coefficients and the two residual bounds were tuned once on wind seeds 1–2 by J and are the same everywhere.

2.4 The supervising agent

Every 30 episodes the policy is evaluated on wind seeds 1–2 and the agent (an OpenAI-compatible reasoning model in JSON mode, about five calls per run) receives the training diagnostics, the evaluation, the decision history with outcomes and the objective’s definition. It returns $K = 3$ candidate *reward expressions*—sandboxed Python over (region, wind label, speed error, power ratio, both fatigue proxies, actuation) that replace (2), in the manner of Eureka [10]. Each candidate is trained for a 30-episode fork from the same checkpoint, the forks are evaluated on wind seeds 1–2, and the best fork continues (*fork verification*); a guardrail rolls the run back to its best checkpoint when the objective drops by more than five points or energy loss exceeds 1 %. The controls used in the ablations run through the same loop: the fixed reward (2) with no agent; the agent asked once at the first decision and applied without a fork; and a random *structural* search over expressions drawn from a fixed grammar of the agent’s own term shapes with log-uniform coefficients.

Table 1: The supervised residual on the MPC base (held-out; mean $\pm$ s.d. over $n = 3$ seeds; composition on wind seeds 3–6 as power MSE / speed MSE / tower fatigue / blade fatigue, % reduction vs. GSPI; difference to the MPC alone with wins on wind seeds 3–6 / 7–10). Energy is within ± 0.2 % of GSPI in every row.

controller	n	J seeds 3–6	J seeds 7–10	$P/\omega/T/B$ (seeds 3–6)	ΔJ vs. MPC alone (wins)
LPV-MPC alone, exact model	—	16.28	17.22	11.0 / 22.3 / 24.1 / 7.8	—
MPC + agent-supervised residual	3	17.23 ± 1.30	18.84 ± 1.04	13.4 / 24.7 / 23.6 / 7.2	+1.0 / +1.6 (2/3, 3/3)
MPC + residual, fixed reward (ablation)	3	16.60 ± 0.78	18.06 ± 1.10	12.1 / 23.6 / 24.2 / 6.4	+0.3 / +0.8 (2/3, 2/3)
LPV-MPC alone, $C_P/C_T \times 0.95$ in its model	—	8.53	3.57	1.5 / 4.7 / 23.4 / 4.5	—
$\times 0.95$ MPC + agent-supervised residual	3	12.66 ± 5.13	18.25 ± 2.52	13.0 / 22.7 / 9.3 / 5.7	+4.1 / +14.7 (2/3, 3/3)
$\times 0.95$ MPC + residual, fixed reward (ablation)	3	14.17 ± 0.82	15.35 ± 0.68	18.6 / 27.6 / 6.7 / 3.8	+5.6 / +11.8 (3/3, 3/3)

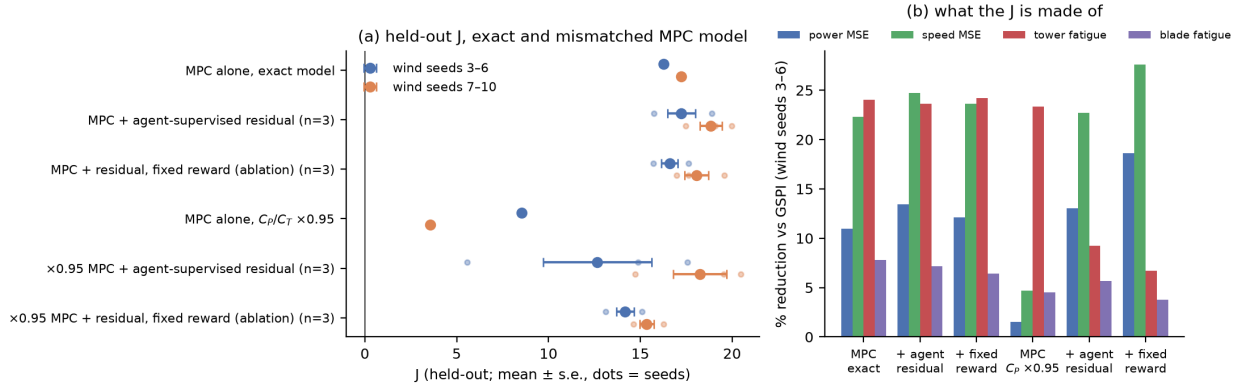


Figure 2: The residual on the MPC base. (a) Held-out J on both wind sets for the MPC alone and with the residual (agent-supervised and fixed reward), with an exact model and with the power/thrust coefficients 5 % low in the MPC’s model (mean $\pm$ s.e., dots are seeds). (b) The four terms of J on wind seeds 3–6 for the same rows.

3 Results

3.1 The supervised residual on the MPC

Table 1 and Fig. 2 give the main result. On the exact-model MPC the agent-supervised residual adds +1.0 / +1.6 on the two held-out sets, above the MPC in five of six seed-set pairs; the sixth is a seed whose best checkpoint on the supervisor winds is the initial policy, i.e. the MPC itself (−0.5). All four metrics improve in every seed; the composition is the MPC’s—a strong regulation gain and a 24 % tower-fatigue reduction—with regulation improved further. The MPC leaves the residual little room: with a zero residual the controller is already positive on every metric, and on the supervisor winds several runs never exceeded their initial policy.

3.2 The residual repairs the model

When the MPC’s power and thrust coefficients are 5 % low in its model, the controller alone drops to 8.5 / 3.6: it under-pitches for the plant it actually controls, its speed sits above rated for tens of seconds at a time, and its regulation terms on wind seeds 7–10 are worse than GSPI’s. The supervised residual restores 12.7 / 18.3, on wind seeds 7–10 above the exact-model MPC alone, in every seed but one on wind seeds 3–6 (that seed’s policy generalised poorly from the supervisor winds, 5.6 against a supervisor-wind 19.6). The residual has learned the part of the plant the model gets wrong, on wind realisations it never saw in training. The composition changes accordingly: the mismatched MPC’s remaining strength was tower fatigue and its weakness regulation, and the residual buys regulation back at a cost in tower fatigue (23 % to 9 %), landing at a balanced controller.

3.3 One held-out episode

Fig. 3 (left column) shows the repair in time on one held-out 15 m/s episode chosen by a stated rule: the mismatched MPC alone lets power ride above rated during the gust at 50–80 s and drifts around it afterwards (power MSE 3.1×10^{-5}); with the residual the power stays on the reference (1.7×10^{-5} supervised, 2.0 fixed) at a modest cost in tower-base moment. The two residuals are indistinguishable by eye.

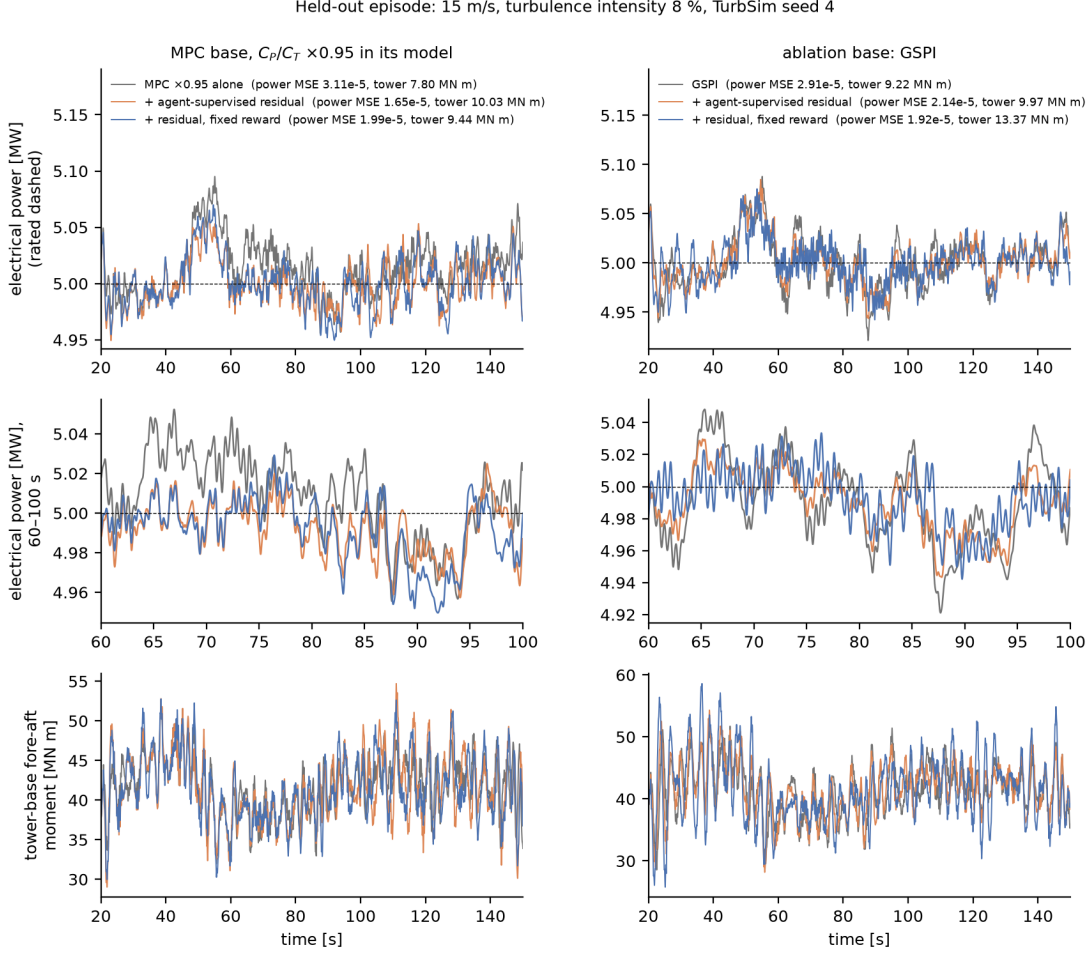


Figure 3: One held-out above-rated episode: how close each controller holds power to rated, and what it costs the tower. Rows: electrical power over the scored window (rated dashed); the same, 60–100 s; tower-base fore-aft moment. Left column: the MPC with the 5 % model error alone and with the agent-supervised and the fixed-reward residual. Right column: the ablation base (§4), GSPI with the two residuals (one seed each, the seed nearest its arm’s mean). The legend gives the episode’s power MSE on above-rated steps and its tower-base fatigue; above rated the generator torque is constant, so the generator-speed trace is the power trace scaled. The episode is the 15 m/s realisation on which the two GSPI-base arms’ tower-fatigue difference is closest to its mean over the four held-out realisations.

3.4 What the agent writes

The rewards the agent keeps have a structure that no coefficient of (2) can produce. The best-checkpoint expression of the strongest MPC-base run (held-out 18.9/20.0) is, in the sandbox’s variables (region = router region, region_w = wind label, d_wg = relative speed error, p_ratio = power over paired GSPI power, load_t, load_b = fatigue proxies, act = actuation),

$$\begin{aligned}
 & (90 \cdot \tanh(p_ratio - 1) \text{ if } region == 0 \text{ else } 0) \\
 & - (24 \cdot \tanh(\text{abs}(d_wg) / .005) + 16 \cdot \tanh(\text{abs}(p_ratio - 1)) \text{ if } region_w == 1 \text{ else } 0) \\
 & - \tanh(load_t) \cdot (.85 \text{ if } region == 0 \text{ else } .35) - .8 \cdot \tanh(load_b) - .07 \cdot \tanh(act)
 \end{aligned}$$

Three departures from (2) recur across the agent’s MPC-base runs (Appendix E): the regulation term on the above-rated subset is *saturated* ($\tanh |\delta\omega|/0.005$, or its square) instead of quadratic, so that a large error in a gust does not dominate the return; a second saturated term penalises *power* deviation on the same subset, which (2) does not contain although power MSE is half of the objective’s regulation score; and the fatigue terms are *bounded*, $\tanh \ell$, in several runs *centred* at the baseline’s level, $\tanh(\ell - 1)$, so that load below the baseline’s is not paid for. The ablations that follow show that these shapes are what the agent contributes.

Table 2: The ablation base: the same residual and the same loop on the GSPI base, where the residual is the whole improvement (held-out J , mean $\pm$ s.d.; composition on wind seeds 3–6; “balanced” counts seeds on which all four metrics improve; difference to the fixed reward with wins and exact p on wind seeds 3–6 / 7–10).

arm	n	J seeds 3–6	J seeds 7–10	$P/\omega/T/B$	balanced	ΔJ vs. fixed (wins, exact p)
fixed reward (tuned), no agent	5	4.86 ± 2.76	5.50 ± 2.09	$17.4 / 19.0 / -17.6 / 0.7$	0/5	—
agent-written reward, verified loop	5	5.30 ± 4.17	6.04 ± 4.28	$6.8 / 10.1 / +1.5 / 2.7$	3/5	$+0.4 / +0.5$ (3/5, 0.75 / 0.69)
agent asked once, no verification	5	4.97 ± 1.90	6.66 ± 2.75	$11.6 / 13.3 / -5.5 / 0.5$	1/5	$+0.1 / +1.2$ (2/5, 3/5; ≈ 1)
random structure from the agent’s vocabulary, verified loop	3	6.13 ± 0.52	7.06 ± 1.35	$9.5 / 10.5 / +2.1 / 2.5$	1/3	$+0.4 / +0.3$ (2/3, 1/3; ≈ 1)

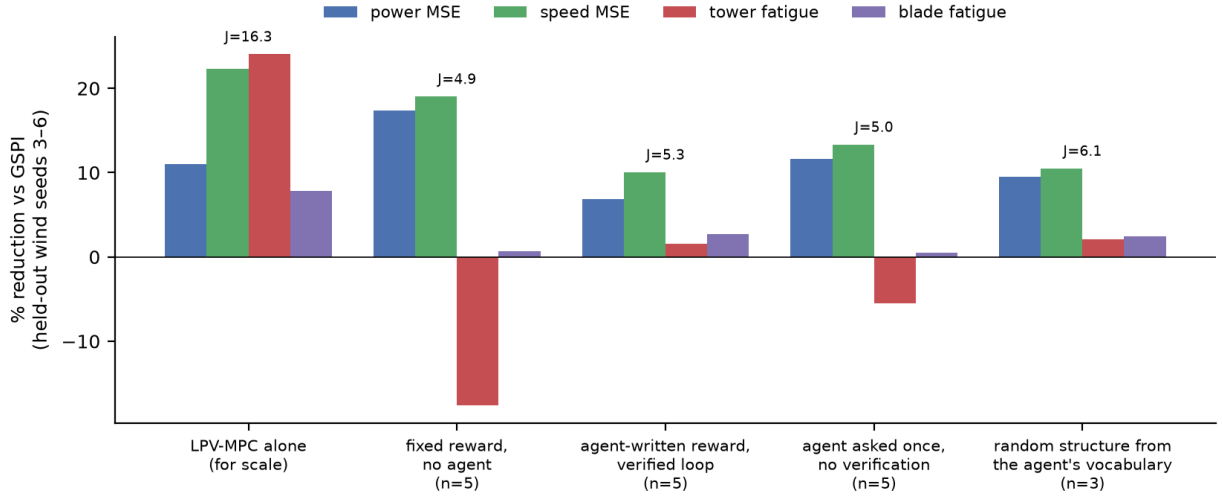


Figure 4: What each ablation arm’s held-out J is made of on the GSPI base (wind seeds 3–6, arm means; the arms of Table 2), with the MPC alone for scale. The fixed reward and the agent asked once buy regulation with tower fatigue; the verified agent-written reward and the verified random draw from its vocabulary have every term positive.

4 Ablations: what the agent contributes, and what it does not

Is the agent’s reward needed on the MPC base? Not for the level. The fixed reward (2), tuned once, reaches $16.60 \pm 0.78 / 18.06 \pm 1.10$ on the exact-model MPC against the agent’s $17.23 \pm 1.30 / 18.84 \pm 1.04$: a difference of $+0.6 / +0.8$ in two of three seeds ($p = 0.75$ at $n = 3$), inside the seed spread. On the mismatched MPC the fixed reward is $14.17 \pm 0.82 / 15.35 \pm 0.68$ against the agent’s $12.66 \pm 5.13 / 18.25 \pm 2.52$, one poor agent seed deciding the first set. On this base the residual has little to do and the reward’s shape has little to change; the agent’s effect on the level of the objective is within noise at these seed counts.

What the agent changes where the residual has more to do. On the GSPI base the residual is the whole controller’s improvement, and the reward’s shape decides what that improvement is made of (Table 2, Fig. 4). The fixed reward buys regulation with tower fatigue (-17.6%), the opposite of what a residual load controller is for. The agent’s rewards are the only arm whose mean tower fatigue improves and the only one with balanced seeds: 3 of 5, against 0 of 5 for the fixed reward. Their level is not higher ($+0.4 / +0.5$, $p = 0.75 / 0.69$); their composition is different, and Fig. 3 (right column) shows what that looks like on one episode: both residuals hold power closer to rated than GSPI, the fixed reward by driving the tower harder (13.4 against 9.2 MN m), the agent’s without (10.0).

Verification makes it real; the vocabulary is the prior. Asked once at the first decision and applied without a fork, the same agent is the fixed reward (1 of 5 balanced). A random search over expressions drawn from a fixed grammar of the term shapes the agent uses—the saturated regulation forms, the centred bounded load forms, log-uniform coefficients—verified by the same loop, lands where the agent lands: $6.13 \pm 0.52 / 7.06 \pm 1.35$, positive mean tower fatigue, the balanced solution in 1 of 3 seeds, the saturated forms kept. Whether the agent finds the balanced solution more reliably (3/5 against 1/3) is not separable at these seed counts. What the agent demonstrably contributes is the vocabulary—it wrote, unprompted, the shapes the grammar of the control was built from—and what makes the vocabulary count is verification against the objective, not the agent’s search and not its prompt.

5 Discussion and limitations

Put learning where the model fails: on the exact-model MPC the residual adds about a point and repairs nothing, on the mismatched MPC it repairs 4 to 15. Put the agent on the structure of the reward and verify what it writes against the objective you report: its level effect is within noise on both bases, its structural effect is real, and it survives only inside the verification loop. Evaluate the reference controller as itself: the channel through which the MPC was first evaluated cost it four points, more than any agent moved anything.

Seed budgets are three for the MPC-base arms and five for the core GSPI-base arms; no agentic difference in level is significant, and the composition effect (3/5 balanced) rests on few seeds. One plant, one training turbulence intensity, two held-out sets of four realisations, one reasoning model with one prompt family. The MPC has no preview, a crude tower-state estimator and a two-second horizon; the model error studied is a scalar scaling of the aerodynamic coefficients, not an identified model. The residual policy observes the simulator’s hub wind where the MPC uses ROSCO’s estimate. An individual-pitch extension of the same supervisor was attempted and did not win on any of its levers; the boundary of the present claims is collective pitch.

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A Objective, metrics and the tuned fixed reward

The four metrics of (1) are computed per episode against the paired GSPI run on the same wind realisation: power MSE and generator-speed MSE relative to rated on the steps where hub wind exceeds rated (the 8 m/s episodes contribute no regulation term), and damage-equivalent loads of tower-base fore-aft moment and blade-root out-of-plane moment over all episodes. Each episode’s reduction is clipped to $\pm 100\%$ before averaging so that a single blown regulation episode cannot own the objective. Energy is the total over all episodes.

The tuned fixed reward (2) is the per-knob geometric median of the best checkpoints of a three-seed random-search arm over its six knobs on the GSPI base (fork-verified, J -selected on wind seeds 1–2): $w_P = 220$, $w_\omega = 25.2$, $\lambda_T = 1$, $\lambda_B = 1.31$, residual bounds 0.074 rad (below rated) and 0.05 rad (above). That search raised held-out J from $4.60 \pm 2.38 / 5.96 \pm 2.02$ (untuned) to $6.70 \pm 2.12 / 7.89 \pm 2.21$ but never kept a candidate with $\lambda_T > 1$: fatigue responds over tens of episodes, regulation within one fork, so a 30-episode verification horizon is myopic for that knob (Fig. 6b). A sweep of the tower weight from the tuned default (seed 0, wind seeds 3–6 / 7–10) confirms the balanced solution is not on that axis: $\lambda_T = 1$: 5.0 / 5.9; 3: 1.4 / 3.7; 10: $-2.9 / -3.3$; 30: the policy never left its initial state. The same default is used unchanged on the MPC base.

B Further supervision arms on the GSPI base

The supervision loop of §2.4 was also run with the agent on two other levers, each with a random-search control, and with two variants of the reward-expression agent (Table 3, Fig. 5). *Hyper-parameters*: the agent proposes PPO learning rates, GAE λ , entropy and clip; control: random search. *Combined*: one agent proposes hyper-parameters and a reward expression together. *Told a different objective*: the reward-expression agent is told a load-priority objective (tower fatigue first, speed regulation as a penalised constraint) while J still selects every fork and checkpoint.

Table 3: Further arms on the GSPI base (held-out J , mean $\pm$ s.d.; composition on wind seeds 3–6; balanced seeds; difference to the tuned fixed reward with wins and exact p on wind seeds 3–6 / 7–10).

arm	n	J seeds 3–6	J seeds 7–10	$P/\omega/T/B$	balanced	ΔJ vs. fixed (wins, exact p)
LLM hyper-parameters	3	5.26 ± 2.39	7.13 ± 3.05	$17.1 / 20.7 / -18.1 / 1.2$	0/3	$-0.5 / +0.3$ (2/3, 1.0)
random hyper-parameters	3	4.93 ± 2.02	5.96 ± 2.17	$15.1 / 17.0 / -13.2 / 0.8$	0/3	$-0.8 / -0.9$ (2/3, 1.0)
LLM hyper-parameters + reward, one agent	5	6.37 ± 2.94	8.65 ± 2.60	$15.9 / 16.3 / -7.6 / 0.9$	0/5	$+1.5 / +3.2$ (4/5, 0.31 / 0.13)
LLM reward, told a load-priority objective	5	7.00 ± 2.12	8.33 ± 2.38	$9.5 / 14.6 / +0.9 / 3.0$	3/5	$+2.1 / +2.8$ (4/5, 0.31 / 0.19)

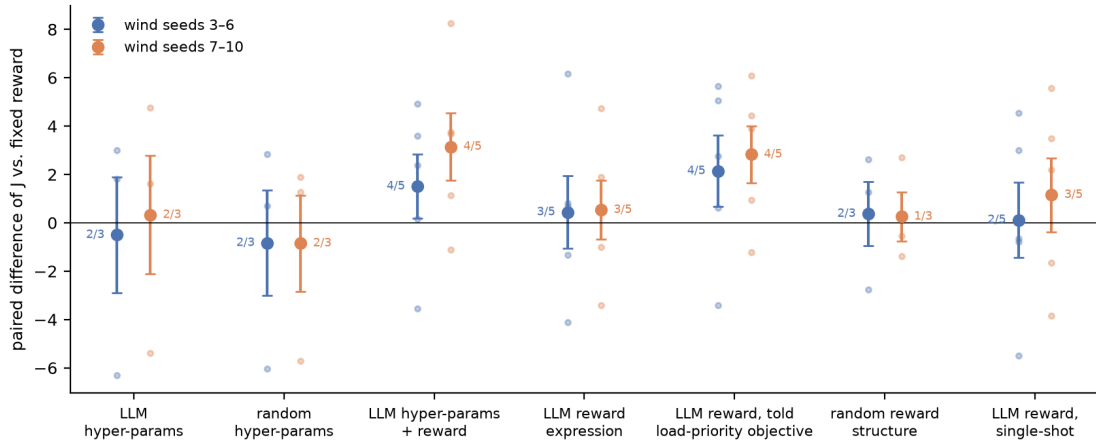


Figure 5: Seed-paired difference of held-out J between each supervised arm and the tuned fixed reward on the GSPI base, both held-out wind sets (mean $\pm$ s.e., dots are seeds, labels are wins/ n). No arm is significantly above zero.

The hyper-parameter agent is worth $-0.5 / +0.3$ and random hyper-parameter search $-0.8 / -0.9$: the two proposers are indistinguishable and neither moves the tuned fixed reward; on an earlier, defective default (Appendix C) both had appeared valuable. The combined agent raises the level ($+1.5 / +3.2$, 4/5) but never the composition (0/5 balanced): it buys regulation the way the hyper-parameter arms do. The agent told a load-priority objective writes the same shapes

as the agent told J and, with J selecting its forks, reaches the balanced solution as often (3/5) with the largest paired gain of any arm (+2.1/+2.8, 4/5); its fork records show it choosing between candidates whose told objective and J disagreed, and J won each time because J selects. Pooled over the two looped reward-expression arms the balanced solution appears in 6 of 10 seeds, in 1 of 5 without the loop and in 0 of 5 fixed-reward seeds.

C Why the reward’s speed term is gated by the wind label

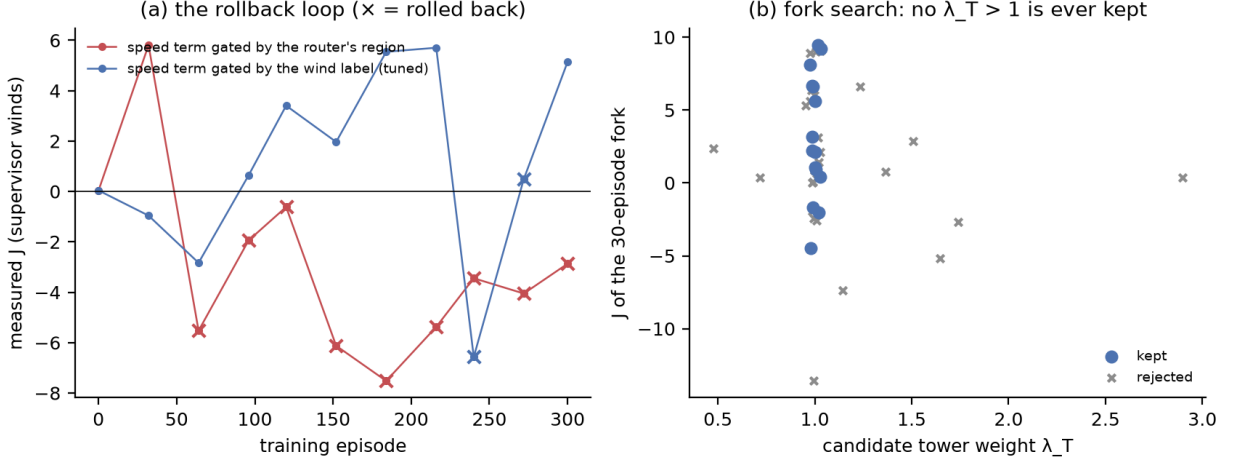


Figure 6: (a) Measured J on the supervisor winds at every decision of a fixed-reward run whose speed term is gated by the router’s region (red) and of the tuned default gated by the wind label (blue); crosses mark rollbacks. (b) Every fork candidate’s tower weight against the J of its 30-episode fork in the weight search; no candidate with $\lambda_T > 1$ is ever kept.

The objective’s regulation metrics are scored on the steps where hub wind exceeds rated, whatever the controller does. If the reward’s speed term is instead gated by the router’s region, the transition steps at 12.5 m/s—router “below rated”, objective “above”—carry no speed penalty, and the policy is rewarded there for load reduction that the objective scores as regulation loss. Under that gating every fixed-reward run found an early optimum and then rolled back at every later decision (Fig. 6a): held-out J $1.59 \pm 0.87 / 0.97 \pm 1.37$ ($n = 3$). On that default, hyper-parameter supervision appears valuable (+5.7/+8.0 for the LLM, +5.8/+7.5 for random search, both by lengthening the credit window, GAE λ $0.98 \rightarrow 0.995$, and slowing the actor). Gating the speed term by the wind label removes the loop without any agent, and on that default the same hyper-parameter arms are worth nothing (Appendix B). The aligned gating is used throughout.

D The MPC: cost scaling, selection, model error

The controller model is $J\dot{\omega} = T_{\text{aero}}(v - \dot{x}, \omega, \beta) - T_{\text{gen}}(\omega)$, $m\ddot{x} = F_{\text{thrust}}(v - \dot{x}, \omega, \beta) - c\dot{x} - kx$ with the first tower fore-aft mode at 0.324 Hz, modal mass 437 t and structural damping ratio 0.01, linearised at the operating point from the C_P/C_T tables every 0.1 s and discretised exactly; tower states are estimated by leaky double integration of the measured nacelle acceleration; the wind input is ROSCO’s estimate low-passed at 0.35 rad/s; torque is native ROSCO. The condensed QP over $N = 20$ steps minimises $\sum_k q (\delta\omega_k / (0.005 \omega_{\text{rated}}))^2 + q_t (\dot{x}_k / 0.2)^2 + r (\Delta\beta_k / 0.002)^2$ subject to the live pitch floor, the pitch limit and the rate limit. Selection on wind seeds 1–2 by J over $N \in \{20, 40\} \times r \in \{0.1, 0.3, 1, 3\} \times q_t \in \{0, 0.1, 0.3, 1, 3\} \times$ two wind-filter corners (82 points) gives $N = 20$, $r = 0.3$, $q_t = 3$; $N = 40$ is worse everywhere. The regulation-only cost of the published verdict (references 1 and 0.1 rad, $q_t = 0$, $r = 0.02$) scores $-2.19 / -6.20$ as the base controller with tower fatigue -22% . Table 4 collects the held-out rows, the model-error study of §2.2 and off-design winds; every row is the MPC as the base controller with a zero residual, and the model error is applied to the MPC’s internal model only.

Table 4: LPV-MPC as the base controller (J -selected weights) on the held-out sets, and under perturbation of its internal model only and off-design winds (wind seeds 3–6 unless noted; the off-design rows use seeds 3–4).

variant	J seeds 3–6	J seeds 7–10	$P / \omega / T / B$ (seeds 3–6)	note
exact model	16.28	17.22	11.0 / 22.3 / 24.1 / 7.8	base of §3.1
$C_P/C_T \times 0.95$	8.53	3.57	1.5 / 4.7 / 23.4 / 4.5	base of §3.2
regulation-only cost ($q_t = 0$, unscaled references)	−2.19	−6.20	10.5 / 1.2 / −22.0 / 1.6	the published verdict
$C_P/C_T \times 0.85 / \times 1.05 / \times 1.15$	−24.0 / 5.7 / −24.6			regulation collapses at $\pm 15\%$
tower frequency $\times 0.9 / \times 1.1$	12.0 / 13.4			
modal mass $\times 0.8 / \times 1.2$	13.3 / 17.6			
turbulence intensity 14 % / 22 % (15 m/s) / 18 m/s	12.6 / 14.2 / 23.1			off-design winds

E Reward expressions written by the agent

Best-checkpoint expressions of the agent on the MPC base, variables as in §3.4. Exact model, seed 1 (held-out 17.1/19.1):

```
(55*tanh(3*(p_ratio-1)) if region==0 else 0)
- (14*tanh(3*abs(p_ratio-1))**2 if region_w else 0)
- (16*tanh(abs(d_wg)/.005)**2 if region_w else 0)
- .8*tanh(load_t) - .8*tanh(load_b) - .12*tanh(act)
```

$C_P/C_T \times 0.95$, seed 1 (held-out 17.6/19.5):

```
(230*tanh(4*(p_ratio-1)) if region==0 else -20*tanh(5*(p_ratio-1))**2)
- (25*tanh(d_wg/.005)**2 if region_w else 0)
- (1.3 if region==0 else .35)*tanh(load_t)
- (.7 if region==0 else 1.2)*tanh(load_b) - .08*tanh(act)
```

$C_P/C_T \times 0.95$, seed 2 (held-out 14.9/14.7), best checkpoint; the same run’s later decisions switch to centred fatigue terms $\tanh(\ell - 1)$:

```
100*(1-region)*tanh(3*(p_ratio-1)) - 32*region_w*tanh((d_wg/.005)**2)
- 24*region_w*tanh(14*(p_ratio-1)**2)
- (.8+.2*region)*tanh(load_t) - .7*tanh(load_b) - .1*tanh(act)
```

On the GSPI base (the ablation base of §4), seed 0:

```
180*tanh(15*(p_ratio-1)) if region==0
-40*sqrt(tanh(abs(d_wg)/.005)**2 + tanh(60*abs(p_ratio-1))**2) if region_w==1
-.9*tanh(log((2+load_t+load_b)/4)) - .1*tanh(act)
```

Seed 1, told the load-priority objective (Appendix B):

```
155*tanh(p_ratio-1) if region==0
-26*tanh(abs(d_wg)/.005) if region_w==1
-(1.4 if region==0 else .5)*tanh(load_t)
-(1.0 if region==0 else 1.5)*tanh(load_b) - .1*tanh(act)
```

Seed 0, told the load-priority objective:

```
130*tanh(6*(p_ratio-1))*(region==0) - 24*tanh(d_wg/.005)**2*(region_w==1)
+ 1.0*tanh((1-load_t)/(1+load_t)) + 1.1*tanh((1-load_b)/(1+load_b)) - .04*tanh(act)
```

The random structural control samples expressions of the same families with log-uniform coefficients spanning the ranges the agent used. Speed terms, in $x = \delta\omega/0.005$: x^2 , $|x|$, $\tanh|x|$, $\tanh(x^2)$, $\sqrt{|x|}$. Load terms, in $\ell \in \{\ell_T, \ell_B\}$: ℓ , $\min(\ell, 3)$, $\tanh(\ell - 1)$, $\tanh \ell$, $\max(0, \ell - 1)$, $\log(1 + \ell)$. An optional power-deviation term on the wind-labelled subset.

F Training-time behaviour and the full controller list

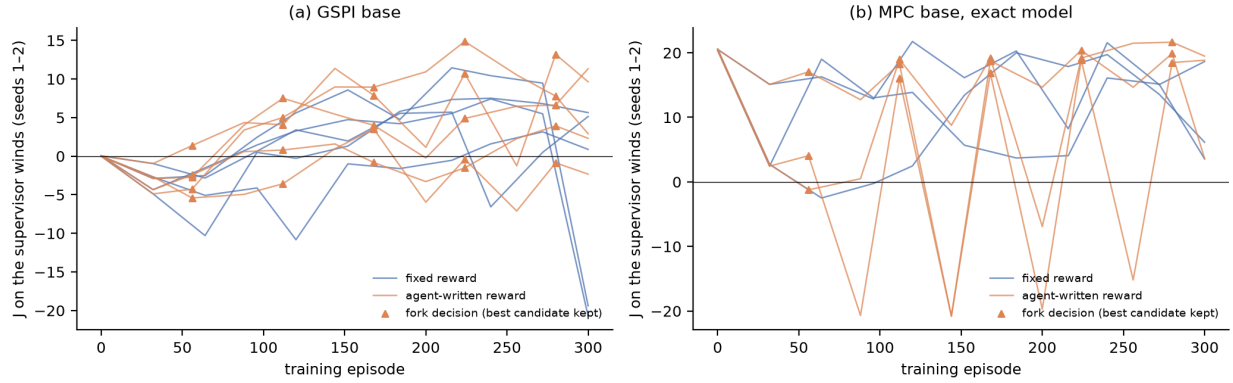


Figure 7: J on the supervisor winds at every evaluation during training, fixed-reward and agent-reward runs, (a) on the GSPI base and (b) on the MPC base with the exact model; triangles mark fork decisions of the agent runs, and the deep dips of the agent runs on the MPC base are evaluations that triggered the guardrail’s rollback. On the MPC base the episode-0 policy is the MPC itself and half of the runs never exceed it.

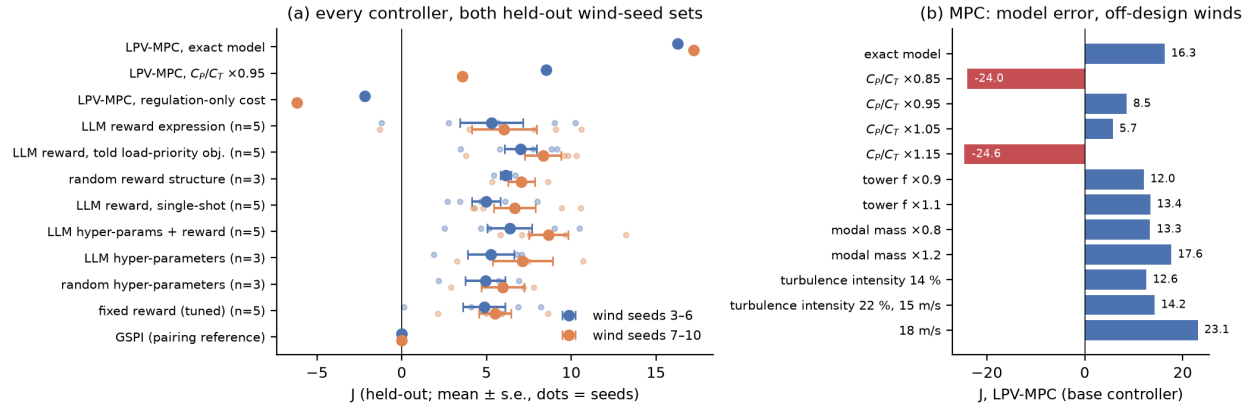


Figure 8: (a) J of every GSPI-base controller and of the MPC (as base controller) on both held-out sets. (b) The MPC under model error and off-design winds (the data of Table 4).

G Compute and reproducibility

One 300-episode training run takes 65–85 minutes on 16 CPU cores (8 OpenFAST workers, two runs in parallel); the MPC in the worker adds 3.6 ms per 0.1 s solve. A supervised run makes about five LLM calls. All runs write rolling checkpoints and can be paused and resumed; every number in this paper is regenerated by the scripts in the repository from the run artefacts, and the figures by a single script.